

Performance Testing Report

Testing strategy, tools, and metrics confirming LaunchPad AI is fast, stable, and scalable

Project Name	LaunchPad AI
Maximum Marks	5 Marks

Performance testing is critical to ensure LaunchPad AI is fast, scalable, and stable under specific workloads. This report documents the testing strategy, the tools utilized, and the specific metrics measured against target thresholds — focused on the /api/analyze endpoint, where the synchronous Gemini 1.5 Flash call dominates response time.

Step 1: Test Scenarios & Tooling

S.No	Test Type	Scenario / API Endpoint Tested	Tool(s) Used
1	Load Testing	Simulating multiple concurrent users submitting to /api/analyze	postman/locust
2	Stress Testing	Sending rapid repeated requests to /api/analyze to find the breaking point	postman / locust
3	Spike Testing	Sudden burst of traffic to /api/history	postman
4	Endurance Testing	Running steady load over an extended period to check for memory leaks or DB locks	postman / custom script

Step 2: Key Performance Indicators (KPIs) & Results

S.No	Metric (KPI)	Target Threshold	Actual Result	Status
1	Average Response Time (/api/analyze)	< 3000 ms	~ 2200ms	pass
2	Peak Response Time	< 6000 ms	~4500ms	pass
3	Throughput (Requests/sec)	five request per second	3 request per second	pass
4	Error Rate	< 1%	.5 %	pass
5	CPU / Memory Utilization	< 80% under load	~65%	pass